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P1. when the ball is going upwards:

$$a = -\frac{mg + kv}{m} \quad \frac{dv}{dt} = -g - \frac{k}{m}v = -\frac{k}{m}\left(v + \frac{mg}{k}\right)$$

$$\int_{v_0}^{v(t)} \frac{dv}{v + \frac{mg}{k}} = \int_0^t -\frac{k}{m} dt$$

$$\ln\left|v(t) + \frac{mg}{k}\right| - \ln\left|v_0 + \frac{mg}{k}\right| = -\frac{k}{m}t$$

$$\frac{v(t) + \frac{mg}{k}}{v_0 + \frac{mg}{k}} = e^{-\frac{k}{m}t}$$

$$v(t) = e^{-\frac{k}{m}t} \left(v_0 + \frac{mg}{k}\right) - \frac{mg}{k}$$

When $v(t) = 0$

$$\frac{dx(t)}{dt} = v(t)$$

$t_h = -\frac{m}{k} \ln \frac{mg}{kv_0 + mg}$ at the highest point.

$$\int_0^{x(t)} dx(t) = \int_0^t v(t) dt$$

$$x(t) = -\frac{m}{k} \left(v_0 + \frac{mg}{k}\right) e^{-\frac{k}{m}t} - \frac{mg}{k}t + \frac{m}{k} \left(v_0 + \frac{mg}{k}\right)$$

$$\text{When } t = t_h = -\frac{m}{k} \ln \frac{mg}{kv_0 + mg} \quad v(t) = e^{-\frac{k}{m}t_h} \left(v_0 + \frac{mg}{k}\right) - \frac{mg}{k} = 0$$

$$x(t_h) = -\frac{m}{k} \times \frac{mg}{k} + \frac{mg}{k} \times \frac{m}{k} \ln \frac{mg}{kv_0 + mg} + \frac{m}{k} \left(v_0 + \frac{mg}{k}\right)$$

$$= \frac{m^2}{k^2} g \ln \frac{mg}{kv_0 + mg} + \frac{m}{k} v_0$$

P2. (a) $x(t) = A \cos(\omega t + \phi) = A \cos(\omega t) \cos \phi - A \sin(\omega t) \sin \phi$

$$\text{so } A \cos \phi = B \quad -A \sin \phi = C$$

$$A = \sqrt{B^2 + C^2}$$

$$\phi = -\arctan \frac{C}{B}$$

P3. $F = \rho g V = -\rho g S x$ given the positive direction of x downwards

$$ma = -\rho g S x \quad \text{so } \ddot{x} + \frac{\rho g S}{m} x = 0 \quad \text{which is the form of simple harmonic motion}$$

$$\frac{\rho g S}{m} = \omega^2 = \left(\frac{2\pi}{T}\right)^2 \quad \text{so } \rho = \left(\frac{2\pi}{T}\right)^2 \frac{m}{g S} = \frac{4\pi^2}{T^2} \frac{W}{g^2 S}$$

P4 for both the platform and the block, take the ground as FOR.

$$\ddot{x} + \omega^2 x = 0$$

$$x = A \cos(\omega t + \phi)$$

when the platform reaches its highest point, its acceleration reaches

the highest.

$$\text{When } x = A \quad \ddot{x} = -\omega^2 A$$

to make sure the block keeps in contact with the platform

$$W_0 A \leq g. \quad \text{so} \quad W_0 \leq \frac{g}{A}$$

P5. for critical damping

$$x(t) = D_1 e^{-\frac{b}{2m}t} + D_2 t e^{-\frac{b}{2m}t}$$

$$\text{when } x(t) = 0 \quad D_1 e^{-\frac{b}{2m}t} + D_2 t e^{-\frac{b}{2m}t} = 0$$

$$(D_1 + D_2 t) e^{-\frac{b}{2m}t} = 0 \quad \text{since } e^{-\frac{b}{2m}t} \neq 0 \quad \text{so } D_1 + D_2 t = 0$$

$$\text{so there is at most one solution} \quad t = -\frac{D_1}{D_2}$$

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P6 (a) $A(W_{dr}) = \frac{F_0}{m \sqrt{(W_0^2 - W_{dr}^2)^2 + \left(\frac{b W_{dr}}{m}\right)^2}}$

$$\begin{aligned} (W_0^2 - W_{dr}^2)^2 + \left(\frac{b W_{dr}}{m}\right)^2 &= W_0^4 - 2W_0^2 W_{dr}^2 + W_{dr}^4 + \frac{b^2 W_{dr}^2}{m^2} \\ &= W_{dr}^4 + \left(\frac{b^2}{m^2} - 2W_0^2\right) W_{dr}^2 + W_0^4 \end{aligned}$$

$$\text{when } W_{dr}^2 = \frac{\frac{b^2}{m^2} - 2W_0^2}{-2} = W_0^2 - \frac{b^2}{2m^2} \quad W_{dr} = \sqrt{W_0^2 - \frac{b^2}{2m^2}} \quad A(W_{dr}) \text{ is maximum}$$

(b) $x_s(t) = A \cos(W_{dr}t + \phi)$

$$\tan \phi = \frac{b W_{dr}}{m(W_{dr}^2 - W_0^2)} = -1$$

$$\frac{W_{dr}}{W_{dr}^2 - W_0^2} = -\frac{1}{4} \quad W_{dr} = -2 + \sqrt{4 + W_0^2}$$

(c)
$$\frac{F_0}{m \sqrt{(W_0^2 - \Omega^2)^2 + \left(\frac{b \Omega}{m}\right)^2}} = \frac{F_0}{m \sqrt{(W_0^2 - \Omega_1^2)^2 + \left(\frac{b \Omega_1}{m}\right)^2}}$$

$$(W_0^2 - \Omega_1^2)^2 + \left(\frac{b \Omega_1}{m}\right)^2 = (W_0^2 - \Omega_2^2)^2 + \left(\frac{b \Omega_2}{m}\right)^2$$

$$\text{let } f(\Omega) = W_0^4 - 2W_0^2 \Omega^2 + \Omega^4 + \frac{b^2}{m^2} \Omega^2$$

$$= W_0^4 + \left(\frac{b^2}{m^2} - 2W_0^2\right) \Omega^2 + \Omega^4$$

$$\frac{\Omega_1^2 + \Omega_2^2}{2} = \frac{\frac{b^2}{m^2} - 2W_0^2}{-2}$$

$$2W_0^2 - \frac{b^2}{m^2} = \Omega_1^2 + \Omega_2^2$$

$$W_0 = \sqrt{\frac{\Omega_1^2 + \Omega_2^2 + \frac{b^2}{m^2}}{2}}$$

P7. $W_{dr} = W_0 = \frac{2\pi}{T} = \frac{2\pi V}{l}$

$$W_0^2 = \frac{k}{m} = \frac{2F_0}{m} = \frac{2F_0 g}{W d_0}$$

$$\sqrt{\frac{2F_0 g}{W d_0}} = \frac{2\pi V}{l}$$

the driven frequency must be equal to the natural frequency

$$V = \frac{\lambda}{2\pi} \sqrt{\frac{2F_0 g}{W d_0}} = 0.473 \text{ m/s}$$